Joshua Jason Hanflink

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EDUCATION

North Carolina State University (NCSU) – Raleigh, NC

B.S. in Electrical & Computer Engineering (Double Major) | Expected Dec 2027

- **GPA:** 3.45 | University Honors Program | TRIO SSS-STEM Scholar
- **Relevant Coursework:** Embedded Systems, Complex Digital Systems, Logic Design, Computer Systems Programming, Signals & Circuits, Control Systems, Linear Systems

TECHNICAL SKILLS

- **Languages & Software:** C/C++, Python, MATLAB, Streamlit, Git, Linux/Bash, KLayout, Silvaco TCAD, KiCad EDA, SolidWorks
- **Hardware & Systems:** PCB Design & Layout, Circuit Schematic Analysis, Multimeter & Oscilloscope Diagnostics, Embedded Linux, Raspberry Pi 5, SPI/I2C/UART Protocols, Precision Soldering, DC-DC Power Conversion, High-Voltage Power Systems
- **Semiconductor & Fabrication:** Mask Design (GDSII), Process Simulation, Spin-Coating, E-Beam Evaporation, MA6 Mask Aligner

RESEARCH & PROJECT EXPERIENCE

ASCEND Summer Research Program – 3D Imaging System | *Summer 2026*

- Architected and co-designed electrical schematics for high-speed imaging arrays, implementing a 24V power distribution network using high-efficiency DC-DC buck converters.
- Hand-soldered 6 custom multi-layer perfboard interfaces and conducted rigorous continuity, isolation, and short-circuit testing prior to final system assembly and hardware integration.
- Executed thermal and voltage stress testing under active loads to ensure subsystem reliability and power stability across continuous operational cycles.

ASCEND Summer Research Program – Semiconductor Automation | *Summer 2025*

- Engineered a standalone Python/Streamlit web application automating the conversion of KLayout GDS II mask files into executable Silvaco TCAD simulation scripts.
- Automated oxidation, photolithography, etching, and metal deposition modeling via Victory Process with custom UI parameter controls.

Vertically Integrated Project (VIP) – Foundry at NCSU | *Spring 2025 – Present*

- Operated class-100 cleanroom equipment (spin-coaters, Karl Suss MA6 mask aligners, CHA Mark 40 evaporators) to fabricate semiconductor microstructures and conducted electrical characterization via probe station testing.

Hardware Assembly & Integration - Autonomous Vehicle (ECE 306)

- Assembled and calibrated a TI MSP430-driven autonomous vehicle chassis per mechanical CAD layouts; populated custom PCBs with soldered microcontrollers, optical sensors, and motor driver modules.

LEADERSHIP & INVOLVEMENT

Lambda Theta Phi Latin Fraternity, Inc. – *Recruitment, Scholarship & Social Chair* | *Apr. 2025 – Present*

- Directed recruitment drives and organized cross-functional campus events to enhance fraternity engagement.

Engineering Ambassador & START Mentor – *NCSU College of Engineering* | *Aug. 2024 – Present*

- Hosted technical panels and campus tours for 50+ attendees; mentored first-year engineering students.

Society of Hispanic Professional Engineers (SHPE) – *Member* | *Fall 2023 – Present*

HONORS & AWARDS

Hispanic Scholarship Fund Scholar (2023–2025) | **Congressional Award** (Gold Medal & STEM Star)

Duke of Edinburgh Award USA (Gold) | **Eagle Scout** | **Presidential Volunteer Service Award** (Gold)

Certificates: Google AI Professional, Leadership Development, Rising Leaders, Interdisciplinary Teams